

Technical Document #924: Cybersecurity - Comprehensive Overview

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Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Penetration testing simulates cyber attacks to identify vulnerabilities.
- Vulnerability management identifies, classifies, and remediates security vulnerabilities.
- Incident response plans define how organizations respond to security incidents.